higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 3

(12041033)

11 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 Explain what is meant by *rated current.* (2)

1.2 People, animals and property shall be protected against harmful earth fault currents.

List FOUR protective measures that can be taken against earth fault currents in an electrical installation. (4)

1.3 Cooking appliances include built-in stoves and oven hobs.

State THREE requirements for the installation of a switch disconnector for a cooking appliance. (3)

1.4 An electrical layout plan is part of a building plan that is relevant to an electrician.

Give TWO items that can be obtained from an electrical layout plant. (2)

1.5 You are required to install a light that will automatically switch on at night.

1.5.1 Draw a neat, labelled diagram of a day-night switch controlling one lamp. (6)

1.5.2 Give the recommended circuit breaker size for this sub-circuit. (1)

1.6 Wireways are tubes or troughs that protect electrical conductors and cables.

Name THREE types of wireways. (3)

1.7 A reamer is one of the tools used during the wiring of an installation.

1.7.1 What is the purpose of a reamer?

1.7.2 What could happen if the reamer is not used?

(2 × 2) (4)

[25]

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QUESTION 2

2.1 List THREE components that are found on a domestic distribution board. (3)

2.2 Explain what is meant by the following:

2.2.1 Point of supply

2.2.2 Point of control

(2 × 2) (4)

2.3 It is important that testing and measuring instruments are of good quality in order to arrive at the correct conclusions.

State FOUR qualities of a good test instrument you would prefer using in order to arrive at the correct conclusions. (4)

2.4 You are required to perform the elevated voltage test between incoming neutral and external earth on an electrical installation.

2.4.1 State TWO measures to be followed when conducting this test. (3)

2.4.2 What should you do after obtaining a reading of 53 volts? (2)

2.4.3 What should be the acceptable reading? (1)

2.5 Draw a neat, labelled diagram to show how an insulation resistance test between live and neutral conductors is conducted. (6)

2.6 According to the law, who is responsible for the safety, safe use and maintenance of an electrical installation? (1)

2.7 Who may issue a certificate of compliance in respect of an electrical installation? (1)

[25]

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QUESTION 3

3.1 Troubleshooting is a process of tracing and rectifying faults in machinery and electrical equipment.

List the FIVE-step approach to fault-finding that can be used on any type of electrical circuit. (5)

3.2 You are required to find a fault on a socket outlet circuit installed in the electrical workshop.

Give THREE possible faults that may be found on a socket outlet. (3)

3.3 Diagrams are effective tools that can be used during the fault-finding process.

List THREE types of diagrams used during the fault-finding process. (3)

3.4 Maintenance is a process of keeping electrical equipment in proper working condition.

Explain what is meant by *preventative* maintenance procedures. (2)

3.5 Give TWO reasons why proper planning is an important part of the fault-finding process. (2)

[15]

QUESTION 4

4.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

4.1.1 Low-voltage refers to nominal voltage levels up to and including 10 kV.

4.1.2 Secondary distribution comprises the low-voltage network from which service connections are tapped off to individual consumers.

4.1.3 A ring distribution system is more expensive to build, but is more reliable than a radial distribution system.

4.1.4 In a radial distribution system, failure of a feeder does not interrupt the supply to a substation.

4.1.5 A low-voltage network may be underground, overhead or a combination of both.

(5 × 1) (5)

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4.2 Name ONE material that can be used to manufacture earth electrodes. (1)

4.3 A work permit is a statement that confirms that a circuit or piece of equipment is safe to work on.

State TWO other items that are addressed on the work permit. (2)

4.4 A TN-C-S low-voltage earthing system has got a PEN conductor.

Explain what is meant by a *PEN conductor.* (2)
[10]

QUESTION 5

5.1 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A-H) next to the question number (5.1.1-5.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
5.1.1	Used to correct power factor in a fluorescent lamp	A	reflector lamp
5.1.2	Designed to replace incandescent lamps	B	smaller
5.1.3	Initial cost for incandescent lamps	C	bigger
5.1.4	Can withstand high temperatures and thermal shock	D	low
5.1.5	The size of tungsten halogen lamps compared to standard incandescent lamps	E	capacitor
		F	initial cost is high
		G	compact fluorescent lamp
		H	high

(5 × 1) (5)

5.2 Calculate the maximum permissible volt drop allowed at a socket outlet in an installation if the voltage at the point of supply is 230 V. (3)

5.3 Briefly explain why the mercury vapour lamp cannot be restarted immediately once it has been switched off. (2)

5.4 A preventative maintenance plan should be developed and implemented to keep the lighting system in proper working condition.

List THREE tasks that are done during a preventative maintenance plan. (3)

5.5 Give TWO methods that the municipalities use to switch off geysers during peak periods. (2)
[15]

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QUESTION 6

6.1 Installing solar panels is a high-risk activity and therefore a hazard analysis must be conducted to prevent worksite injuries.

Give TWO benefits of conducting a hazard analysis. (2)

6.2 State the purpose of a blocking diode in a photovoltaic module. (2)

6.3 Give THREE advantages of photovoltaic systems. (3)

6.4 State THREE factors that influence the amount of hot water produced by a solar water heater. (3)

[10]

TOTAL: 100

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